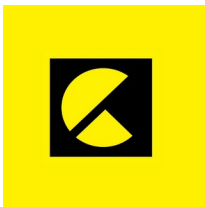


CARPENTERS FIJI



DETAILED PROJECT PLAN

Carpenters Rental, Equipment and Maintenance System

CREMS

A configurable group-wide platform for rental revenue, asset maintenance expenditure and customer service across Carpenters Fiji divisions

Project detail	Information
Client	Carpenters Fiji Pte Limited
Course	CS400 Industry Experience Project - Semester 2, 2026
Assignment	Assignment 1 - Project Plan and Team Charter (15%)
Group number	[INSERT GROUP NUMBER]
Academic supervisors	Amit Kumar; Shanil Naidu
Version and date	Version 2.0 - 7 August 2026

TEAM MEMBERS

Name	Student ID
Kavish Chandra	[INSERT STUDENT ID]
[INSERT TEAM MEMBER 2]	[INSERT STUDENT ID]
[INSERT TEAM MEMBER 3]	[INSERT STUDENT ID]
[INSERT TEAM MEMBER 4]	[INSERT STUDENT ID]

Submission deadline: Sunday 16 August 2026, 11:00 pm Fiji Standard Time

Document control

Version	Date	Prepared by	Purpose
1.0	4 Aug 2026	CREMS project team	Initial rental-focused plan
2.0	7 Aug 2026	CREMS project team	Group-wide modular scope, current requirements and detailed execution baseline

Approval record

Role	Name	Approval/signature	Date
Project leader	[INSERT NAME]		
Client representative	[INSERT NAME]		
Academic supervisor	Amit Kumar		

How to use this plan

This plan is the delivery baseline for the four-person CREMS team. Requirements marked Confirmed may be developed and tested. Requirements marked Proposed or Open require client validation before they become contractual scope. Material changes follow the change-control process in Section 14.

Submission check: Replace every [INSERT ...] field, obtain team charter signatures, update client-approval status, export the complete plan and charter as one PDF, and submit through the project leader account before the deadline.

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1. Executive summary

Carpenters Fiji operates through multiple divisions, including Motors and vehicle rental, Carptract, Hardware, Shipping and Logistics, Property, MH and other group businesses. Rental activity generates revenue; the same asset then requires inspection, preventive servicing and corrective maintenance that generate expenditure. Current and future divisions may rent different asset types, sell services, require quotes, or supply trained personnel with equipment. A single rigid vehicle-rental application would therefore not meet the client requirement for flexibility.

CREMS will be delivered as a configurable, responsive web platform. Shared capabilities - customer accounts, assets, branches, requests, quotes, bookings, agreements, check-out, check-in, inspection, maintenance, invoicing, notifications and reporting - are reusable across divisions. Division, service and workflow configuration determines which capabilities apply. Individual customers receive a simple vehicle-first journey, while corporate customers may access multiple divisions, authorised contacts, job sites, purchase orders and approval rules.

The selected stack is React and TypeScript for the frontend, ASP.NET Core 10 for secure APIs and identity, Entity Framework Core with Microsoft SQL Server for relational data, and Docker for consistent local development. The four-member team will use two-week iterations, continuous integration, peer review and client checkpoints over fourteen weeks from 3 August to 8 November 2026.

Delivery principle: Protect a usable, testable MVP. Configurability and clear extension points are in scope; unconfirmed specialised workflows and external integrations are not silently treated as approved requirements.

1.1 Project at a glance

Dimension	Baseline
Duration	14 weeks: 3 August to 8 November 2026
Team	Four student developers; approximately 6 hours per person per week
Method	Agile delivery in two-week iterations with fortnightly demonstrations
Primary outcome	Responsive group-wide rental and maintenance MVP
Initial operational focus	Vehicle and equipment rental revenue; inspections and maintenance expenditure
Access model	Administrator; division- and branch-scoped Branch Manager and Rental Officer; Individual and Corporate Customer
Cash budget ceiling	FJD 205, subject to approval; expected expenditure may remain zero

2. Domain research and client context

2.1 Carpenters Fiji operating context

Research and the first client meeting indicate that Carpenters Fiji is a group of operating divisions rather than one homogeneous rental business. Divisions share corporate identity and may share customers and locations, but their assets, pricing, safety checks, personnel requirements and commercial workflows differ. The system must therefore separate division data while reusing common capabilities.

Division / area	Observed or proposed activity	CREMS planning implication
Motors / Rentals	Vehicle hire, fleet, parts and service across several locations	Vehicle availability, licence checks, deposits, fuel, damage and branch check-out/check-in.
Carptrac	Heavy machinery, generators, parts and technical service; public rental status requires confirmation	Quote-led equipment workflow, hour meter, transport, trained operator and high-value maintenance controls.
Hardware	Tools, builders equipment and materials; formal hire status requires confirmation	Large categories/SKUs; possible tool-hire module remains configurable and off until confirmed.
Shipping & Logistics	Shipping and logistics services	Quote, job/dispatch, origin/destination and service-status workflow rather than simple daily asset rental.
Property	Residential or commercial leasing and property maintenance	Monthly lease periods, tenant documents, deposits, inspections and work orders.
MH / retail	Retail activity within the group	Represented in group structure; no rental module enabled without confirmed need.

2.2 Rental and maintenance value cycle

1. A customer identifies a need and selects a division or begins with the simple vehicle-rental path.
2. CREMS records an enquiry, availability request or quote depending on service configuration.
3. Staff validate customer eligibility, asset availability, pricing, documents, deposits and approvals.
4. The approved reservation becomes an agreement and check-out record. Revenue is recognised through charges and invoice records.
5. Check-in captures meter, fuel, condition, damage and additional charges.
6. Inspection may return the asset to availability or create maintenance work and expense records.

2.3 Research conclusions

- A modular monolith is appropriate: one deployable application with strong module boundaries is manageable for the team and extensible for the client.
- Division capabilities and workflow rules must be configuration, not scattered conditional code.

- The customer journey requires progressive disclosure so ordinary vehicle customers are not overwhelmed by shipping, property or heavy-equipment choices.
- Personnel-supported hire must distinguish none, optional and required personnel and later support competency verification.
- Carptrac and Hardware rental availability remain client validation questions, not public claims.

3. Problem analysis

3.1 Problem statement

Carpenters Fiji requires a dependable source of truth for rental assets, service requests, customers, agreements, asset condition and maintenance across divisions and branches. Fragmented paper, spreadsheet or isolated processes can cause double booking, incorrect availability, incomplete agreements, weak damage evidence, missed service intervals, untraceable expenditure and limited management visibility. A vehicle-only system would also create expensive redesign when another division is added.

3.2 Information requirements

Information group	Required data
Organization	Division, capabilities, service offerings, branches, division-to-branch participation and operating status.
Identity and access	Staff role, division, branch, account state, customer account linkage, email verification and audit identity.
Customer	Individual or corporate profile, contacts, identification, eligibility, authorised representatives, job sites and rental history.
Asset/service	Asset ID, category, registration/serial, division, branch, status, rate, specifications, personnel requirement and availability.
Commercial transaction	Enquiry/quote, booking period, line items, pricing, tax, deposit, approval, agreement, invoice, payment and balance.
Operations	Check-out/check-in inspection, meter/fuel, photos/evidence, damage, personnel assignment, dispatch and asset transfer.
Maintenance	Job, fault, preventive schedule, supplier, labour/parts, estimated and actual expense, status and return-to-service.
Management	Utilisation, revenue, maintenance expense, overdue rentals, availability, customer activity and exceptions.

3.3 Constraints and assumptions

Type	Constraint or assumption	Planning response
Time	One academic semester and fixed assessment dates	Prioritise demonstrable MVP; time-box discovery and protect scope.
People	Four students balancing other courses	Small work packages, pairing, peer review and visible workload.

Type	Constraint or assumption	Planning response
Requirements	Specialised division rules remain incomplete	Requirements register, prototypes, client decisions and configurable defaults.
Privacy	Customer, identity and agreement data are sensitive	Synthetic development data, least privilege, secure secrets and retention decisions.
Safety	Vehicle/equipment condition and operator suitability affect people and property	Mandatory checks, audit history and no bypass of safety gates without approved authority.
Connectivity	Team uses macOS and Windows; demonstrations may be remote	Docker database, setup guide, repeatable builds and local fallback demonstration.
Technology	Solution must be realistic yet supportable by students	Modular .NET/React/SQL architecture with limited external dependencies.

4. Stakeholders and user groups

Stakeholder	Interest / responsibility	Engagement
Client sponsor / management	Business value, scope, priorities, acceptance and future rollout	Fortnightly checkpoint; milestone acceptance; change decisions.
Division representatives	Correct workflow, terminology, assets, pricing and reports	Focused discovery and prototype review by division.
Branch managers	Branch performance, approvals, staff and exception handling	Workflow validation, UAT and role-specific training.
Rental officers / agents	Fast daily requests, agreements, inspections and returns	Task walkthroughs, usability tests and demonstrations.
Customers	Simple discovery, trusted booking status, clear obligations and privacy	Customer-journey testing using synthetic personas.
Maintenance personnel	Accurate faults, schedules, work status, parts and cost	Maintenance process validation and acceptance scenario.
Carpenters IT / technical owner	Security, deployment, backup, supportability and integration	Architecture/deployment review and handover.
Academic supervisor	Assessment quality, professional conduct and progress	Scheduled updates, plan review and escalation.
Four-person student team	Delivery, learning outcomes, ethical conduct and evidence	Twice-weekly coordination, pull requests and retrospectives.

4.1 User access principles

- Administrators receive group-wide setup and security access.
- Rental officers and branch managers are assigned both a division and a branch; operational records must match both.
- A branch can participate in multiple divisions, but this does not grant a staff member cross-division access.
- Customers use a separate portal and can view only records linked to their verified customer account.
- Corporate authorised contacts and approval limits are an MVP-extension requirement subject to client confirmation.

5. Objectives and success measures

5.1 Objectives

- Deliver one configurable platform that can add a division, branch or service without rebuilding core rental and maintenance functions.
- Prevent conflicting confirmed reservations through authoritative availability and date validation.
- Support traceable request-to-return workflows and convert inspection outcomes into availability or maintenance.
- Separate group, division, branch, staff and customer access using server-enforced authorization.
- Provide a simple customer journey for individuals and a richer cross-division path for corporate customers.
- Make operational and financial information visible through accurate dashboards, reports and audit events.
- Produce a reproducible release, deployment guide, user guidance, UAT evidence and academic report.

5.2 Measurable acceptance criteria

ID	Acceptance measure	Evidence
AC-01	All four user groups sign in and receive correct UI and API permissions; negative access tests pass.	Role/access test matrix
AC-02	Non-admin staff see only operational records matching assigned division and branch.	Cross-scope API tests
AC-03	System rejects an overlapping active/confirmed booking for the same asset and dates.	Automated test and demonstration
AC-04	Vehicle scenario completes request, approval, agreement, check-out, return, invoice and maintenance decision.	End-to-end UAT script
AC-05	Personnel-required equipment cannot proceed without the required assignment/verification once that workflow is enabled.	Equipment UAT scenario
AC-06	Dashboard totals reconcile with underlying transactions and division/branch filters.	Reconciliation checklist
AC-07	Critical status and administration changes record actor, time and material before/after detail.	Audit inspection
AC-08	Core customer and staff workflows pass responsive and plain-language usability checks.	Usability test record
AC-09	Zero open critical/high security or data-integrity defects at handover.	Issue and security registers
AC-10	All members can build and run the documented release on supported environments.	Setup verification checklist

6. Scope baseline

6.1 In scope - MVP

- Configurable divisions, service offerings, capabilities and division-to-branch membership.
- Staff identity, password recovery, administration, audit, role access and division/branch data scope.
- Individual and corporate customer registration, email verification, secure invitation of existing records and account-linked booking history.
- Vehicle and equipment asset register, QR identification, availability, rates, status and personnel requirement.
- Public catalogue, date/branch/division search, enquiry/booking request and reference tracking.
- Booking review, conflict checks, allocation, pricing, deposits, approvals and status transitions.
- Rental agreement, authorised driver, check-out/check-in inspections, evidence, incidents, addenda, invoice and notifications.
- Maintenance jobs, preventive service dates, estimated/actual expense and return-to-service status.
- Corporate account foundation, quotes, purchase-order requirement, job sites and approval records.
- Dashboards, operational reports, task-oriented manager view, email queue and staging-safe delivery controls.
- Deployment/configuration guidance, Postman tests, user guides, UAT, training and handover.

6.2 Out of scope for the assessed MVP

- Live ERP/general-ledger integration or automatic financial posting.
- Production payment gateway and card-data processing.
- Native iOS/Android applications.
- Live GPS/telematics provider integration beyond an extensible data model.
- Payroll, personnel rostering and formal competency/licensing system.
- External marketplace or travel-booking-platform integration.
- Full property-management accounting, shipping manifest/customs integration or retail point-of-sale.
- Production migration of historical or personal data without an approved migration/privacy plan.

6.3 Future options

The architecture will preserve REST boundaries and stable identifiers for approved future modules: payment gateway, ERP/accounting, telematics, document storage, mobile client, electronic signatures, advanced corporate credit, trained-personnel scheduling, shipping dispatch, property tenancy and business intelligence. These are extension options, not implicit MVP promises.

7. Requirements specification

7.1 Functional requirements

ID	Requirement	Priority / status
FR-01	Administrator configures divisions, capabilities, services and active branch participation.	Must / Confirmed
FR-02	Administrator assigns each non-admin staff member one division and one branch.	Must / Confirmed
FR-03	API limits staff operational data by role, division and branch.	Must / Confirmed
FR-04	Customer registers as Individual or Business, verifies email, signs in and views linked history.	Must / Confirmed
FR-05	Staff securely invites an existing customer record to activate online access.	Must / Confirmed
FR-06	Customer searches available assets by dates, branch, type and division without unrelated complexity.	Must / Confirmed
FR-07	System creates reference-traceable requests and prevents conflicting allocations.	Must / Confirmed
FR-08	Staff reviews and confirms customer, asset, period, rate, tax, deposit and approval details.	Must / Confirmed
FR-09	Collection requires applicable agreement, identification, licence/payment and inspection evidence.	Must / Confirmed
FR-10	Return captures condition, meter/fuel, damage and charges; creates invoice and availability decision.	Must / Confirmed
FR-11	Maintenance records planned/corrective work, supplier, parts and estimated/actual expense.	Must / Confirmed
FR-12	Equipment/service configuration records none, optional or required trained personnel.	Must / Confirmed
FR-13	Required personnel assignment and competency evidence gate equipment dispatch.	Should / Proposed
FR-14	Corporate accounts record authorised contacts, job sites, PO requirement, credit terms and approvals.	Should / Partially confirmed
FR-15	Property service supports quote/lease request, monthly period, deposit and inspections.	Could / Proposed
FR-16	Shipping/logistics service supports enquiry, quote and dispatch status.	Could / Proposed
FR-17	Managers receive utilisation, revenue, maintenance expense, overdue and exception views.	Must / Confirmed
FR-18	Critical actions create auditable records and notifications are queued/retried.	Must / Confirmed

7.2 Non-functional requirements

ID	Requirement / target
NFR-01 Security	ASP.NET Identity hashing, secure cookies, HTTPS outside local development, least privilege, server-side policies and secrets outside Git.
NFR-02 Performance	Common list/search APIs should return within 2 seconds at agreed demonstration volume under normal local/staging conditions.
NFR-03 Availability	Staging demonstration can be restored from source, migrations, configuration and backup within 4 hours.

ID	Requirement / target
NFR-04 Usability	Core daily task is discoverable from task-based navigation; representative users complete UAT without developer intervention.
NFR-05 Accessibility	Keyboard operation, visible focus, readable contrast, labels and responsive layouts are checked for core pages.
NFR-06 Maintainability	Typed API contracts, modular domain boundaries, migrations, review and documented configuration.
NFR-07 Data integrity	Unique identifiers, relational constraints, status transitions, concurrency-aware booking checks and audit evidence.
NFR-08 Privacy	Minimum necessary data, synthetic non-production data, restricted access and client-approved retention/export/deletion rules.
NFR-09 Compatibility	Current Chrome/Edge/Safari desktop; responsive support for common tablet/mobile widths; development on macOS and Windows.
NFR-10 Observability	Health endpoint, structured application logs, email delivery status and actionable error responses.

7.3 Core business rules

- BR-01: A confirmed or active booking blocks overlapping periods for the same asset.
- BR-02: An asset in maintenance, out-of-service or retired status cannot be offered as available.
- BR-03: A staff member cannot create or operate on an asset outside both assigned division and branch.
- BR-04: A customer cannot claim an existing record by knowing an email/customer number; staff invitation or approved verification is required.
- BR-05: Vehicle check-out requires applicable identification/licence/payment and signed agreement checks.
- BR-06: Personnel-required services cannot be dispatched without the configured personnel control when enabled.
- BR-07: Return condition determines whether the asset becomes available, enters inspection or generates maintenance.
- BR-08: Original signed agreement snapshots remain immutable; approved changes use addenda.
- BR-09: Financial discounts, waivers and high-value actions follow configured approval thresholds.
- BR-10: Unconfirmed division services remain quote-only, non-public or disabled.

7.4 Open requirements requiring client decision

Decision	Why it matters	Needed by
Formal equipment owner: Carptrac, Hardware or both?	Determines catalogue, workflows, branches and accountable users.	End of W2
Which initial divisions and branches form the pilot?	Controls data collection, UAT scope and staff configuration.	End of W2
When is trained personnel mandatory and who verifies competency?	Defines safety gate, assignment data and legal responsibility.	Before equipment workflow build
Approved rate, deposit, tax, discount and cancellation rules?	Required for accurate pricing and agreement terms.	Before booking UAT
Customer/corporate approval and credit rules?	Determines authorised contacts, PO and manager approval flow.	Before corporate workflow build
Property and shipping MVP expectation?	Determines whether prototypes remain configuration-only or become deliverables.	Feature baseline review
Email sender, retention and direct-delivery approval?	Controls staging/production privacy and notifications.	Before UAT
Deployment owner and target infrastructure?	Controls release architecture, backup and handover.	Before W11

8. Solution design and feasibility

8.1 Alternatives considered

Alternative	Strength	Assessment
ASP.NET Core + React + SQL Server	Enterprise identity/authorization, strong typing, migrations, client-stack alignment and current implementation investment.	Selected: strongest balance of realism, security, scalability and semester feasibility.
Laravel + React + MySQL/SQL Server	Rapid CRUD and approachable PHP ecosystem.	Viable, but rewriting completed identity, API and workflow work introduces schedule and defect risk.
Java Spring + React	Mature enterprise ecosystem and strong typing.	Technically sound but greater setup/learning overhead with no compensating client benefit.
No-code / rental SaaS	Fast initial configuration.	Insufficient control over division rules, audit, data scope, extensibility and academic solution-design outcomes.
Separate system per division	Simple local rules at first.	Rejected: duplicated customers, code and maintenance; poor group reporting and expensive future change.

8.2 Selected architecture

CREMS uses a modular monolith: one frontend, one API and one relational database deployment, with explicit domain modules for identity, organisation, customers, catalogue/assets, booking/rental, maintenance, corporate operations, notifications, administration and reporting. This avoids premature distributed-system complexity while allowing future modules or integrations to use stable APIs.

Layer	Technology	Responsibility
Experience	React 19, TypeScript, Vite, Material UI	Responsive public site, customer portal and task-based staff portal.
Application/API	ASP.NET Core 10 Web API	Validation, authorization, workflow orchestration and REST contracts.
Identity	ASP.NET Core Identity, secure cookies	Staff/customer separation, roles, recovery, verification and session security.
Data	SQL Server 2022, Entity Framework Core	Relational integrity, transactions, migrations, scope queries and reporting.
Delivery	Docker Compose, GitHub, Postman, VS Code	Repeatable environments, collaboration, testing and release evidence.

8.3 Feasibility assessment

Context	Assessment and control	Conclusion
Technical	Core stack already builds; identity, division scope, customer portal, rental lifecycle, QR, maintenance and administration foundations exist. Remaining work is integration, configurable workflows and test depth.	Feasible
Implementation	Four members, modular work packages, peer review and protected MVP boundary fit the semester.	Feasible with scope control
Economic	Open-source/student tools and existing laptops; maximum optional cash spend FJD 205.	Feasible
Aesthetic/usability	Carpenters branding, task-based navigation, responsive UI and user help reduce training burden.	Feasible with UAT
Ethical/privacy	Synthetic data, minimum collection, role scope, audit and client-approved retention reduce exposure.	Feasible with governance
Health and safety	Mandatory condition checks and proposed operator competency gate support safe hire; client rules still required.	Conditional on validation
Societal/cultural	Simple language, individual/corporate journeys and branch-aware service support Fiji-wide users; accessibility and connectivity require testing.	Feasible with inclusive testing
Environmental	Better utilisation/service scheduling can extend asset life; hosting and data retention should remain proportionate.	Positive potential

8.4 Creative and innovative contribution

- Capability-driven divisions allow the same module to support different businesses without copying the application.
- Progressive disclosure presents a simple vehicle journey while retaining cross-division services for corporate users.
- One QR identity supports asset lookup, check-out and check-in with server-enforced scope.
- Rental revenue and maintenance expense are connected through one auditable asset lifecycle.
- Quote-only configuration permits research/prototyping without falsely publishing unconfirmed rental services.

9. Delivery approach and work breakdown

9.1 Agile operating model

The team works in two-week iterations. Each iteration starts with refinement and acceptance criteria, uses small feature branches and peer-reviewed pull requests, and ends with an integrated demonstration. Client decisions are recorded in the requirements register. Features are complete only when authorization, validation, tests, documentation and demonstration data are integrated.

9.2 Work packages

WP	Work package	Outputs / completion evidence
WP1	Discovery, research and governance	Validated scope, project plan, charter, requirements/decision/risk registers and prioritised backlog.
WP2	Architecture and environments	Repository workflow, Docker database, builds, migrations, module boundaries and setup verification.
WP3	Organisation, identity and access	Divisions/branches, staff/customer accounts, recovery/verification, scoped authorization and audit.
WP4	Catalogue, assets and customers	Assets/services, branch/division ownership, customer profiles, corporate foundation and QR identity.
WP5	Requests, quotes and bookings	Public discovery, enquiry/reference tracking, availability, allocation, pricing and approvals.
WP6	Agreement and rental operations	Agreement snapshots, driver/customer checks, check-out, active rental, addenda, check-in and invoice.
WP7	Maintenance and personnel-supported hire	Inspection decision, jobs/cost, preventive schedule and proposed operator assignment control.
WP8	Management information and notifications	Dashboard, reports, email queue, exception tasks and reconciliation.
WP9	Quality, deployment and handover	Automated/manual tests, security/usability checks, UAT, release, training and final report.

9.3 Definition of done

- Requirement and acceptance criteria are linked to the work item.
- Backend and frontend builds pass and migrations are reviewed.
- Authorization is enforced in APIs and tested with allowed and denied users.
- Peer review is complete and no unresolved high-severity comment remains.
- Repeatable tests and demonstration data pass.
- User help, API collection and relevant documentation are updated.

- Client/product-owner acceptance is recorded or follow-up work is scheduled.

10. Timeline, milestones and dependencies

The schedule begins 3 August 2026 and ends 8 November 2026. Activities overlap intentionally: discovery continues through demonstrations, documentation is updated with each feature, and testing begins before feature complete. The Assignment 1 plan and charter baseline is due 16 August 2026.

10.1 Detailed timeline

Weeks / dates	Planned focus	Exit deliverable
W1-2 3-16 Aug	Client research; group/division problem analysis; plan and charter; requirements baseline; environment verification.	M1: submitted plan; prioritised backlog; decisions list.
W3 17-23 Aug	Validate pilot divisions/branches; refine data model; assign existing staff division/branch; API contract review.	Validated organisational and access baseline.
W4 24-30 Aug	Complete customer verification/invitation tests; division/service administration; usability review.	M2: secure account and configuration demonstration.
W5-6 31 Aug-13 Sep	Asset/service catalogue; personnel requirements; corporate profile/contacts; public progressive-disclosure journey.	M3: core records and customer journey accepted.
W7-8 14-27 Sep	Quote/request, availability, pricing, approvals, bookings and conflict testing by division.	M4: request-to-confirmation demonstration.
W9 28 Sep-4 Oct	Agreement templates and configurable collection gates; equipment/personnel proof-of-concept.	Signed agreement and safe check-out demonstration.
W10 5-11 Oct	Return inspection, charges, invoice, maintenance decision and expense reporting.	M5: complete rental lifecycle.
W11 12-18 Oct	Dashboards, revenue/expense reporting, notifications, audit reconciliation and accessibility improvements.	Feature-complete integrated build.
W12 19-25 Oct	Security, authorization, performance, compatibility, backup/restore and regression testing.	Release candidate 1 and evidence pack.
W13 26 Oct-1 Nov	Client UAT, defect correction, user training and deployment rehearsal.	M6: UAT accepted or exceptions recorded.
W14 2-8 Nov	Final release, source, database scripts, documentation, academic demonstration and handover.	M7: handover package and closure report.

10.2 Milestones

Milestone	Target	Acceptance condition
M1 Plan baseline	16 Aug	Plan, charter, scope, requirements, schedule, budget and risks reviewed for submission.
M2 Secure foundation	30 Aug	All members run stack; roles and division/branch negative access tests pass.

Milestone	Target	Acceptance condition
M3 Core records	13 Sep	Divisions, branches, services, assets and customers are maintainable with validation.
M4 Booking workflow	27 Sep	Request-to-confirmation works and conflicting allocation is rejected.
M5 Rental lifecycle	11 Oct	Agreement, check-out, return, invoice and maintenance decision succeed.
M6 UAT complete	1 Nov	Agreed scenarios pass or exceptions are approved and scheduled.
M7 Handover	8 Nov	Release, source, migrations, test evidence, guides and final report delivered.

10.3 Dependency and critical-path view

Activity	Depends on	Critical consequence if late
Scope/decision baseline	Client access and domain research	Rework across model, UI and acceptance tests.
Division/branch configuration	Confirmed pilot structure	Staff cannot be assigned or tested correctly.
Availability/booking rules	Asset and service data	Rental lifecycle cannot be validated.
Agreement/check-out	Confirmed booking and client terms	No legally/operationally credible handover.
Return/maintenance	Active rental and inspection rules	No end-to-end revenue/expense lifecycle.
UAT	Integrated release candidate and scenarios	Handover acceptance delayed.
Deployment/handover	UAT resolution and technical owner	Final delivery incomplete.

10.4 Schedule control

- Track each work item against milestone and requirement ID.
- Review committed versus completed priority work every fortnight; target at least 80%.
- Escalate blockers after two working days.
- Use scope substitution before extending dates: remove equal/greater effort when adding a material requirement.
- Re-baseline only with recorded team, client and supervisor awareness.

11. Resources, roles and responsibilities

11.1 Human resources

Project role	Assigned person	Responsibilities	Effort
Project leader / client liaison	[INSERT NAME]	Plan, meetings, scope/decision control, integration, submission and team coordination.	6 h/week
Backend and data lead	[INSERT NAME]	Domain/API/database, authorization, migrations, backend tests and security.	6 h/week
Frontend and UX lead	[INSERT NAME]	Responsive journeys, accessibility, API integration, UI testing and user help.	6 h/week
Quality and DevOps lead	[INSERT NAME]	Test plan/evidence, Docker/builds, release, backup, documentation and demonstrations.	6 h/week

Roles establish accountability but not silos. Members pair on high-risk features, review each other's changes and rotate meeting-note, demonstration and support duties. Planned team effort is 336 hours (4 people x 6 hours x 14 weeks).

11.2 Technical and information resources

- Four development laptops (one macOS, two confirmed Windows, one team device to be confirmed).
- GitHub repository with issues, feature branches, pull requests and version history.
- .NET 10 SDK, Node.js 22+, React, Material UI, SQL Server 2022, Docker Desktop, VS Code and Postman.
- Client representatives, current forms/agreements, sample asset categories, branch/division information and acceptance decisions.
- Academic supervisor for assessment guidance and escalation.
- Synthetic test dataset; production customer data is not copied into student environments.
- Optional staging host, email sender/domain and secure configuration channel.

11.3 RACI summary

Legend: A = accountable; R = responsible; C = consulted; I = informed.

Activity	Leader	Backend	Frontend	QA/DevOps	Client
Requirements/scope	A	C	C	C	R/C
Architecture/data	C	A/R	C	C	I
Customer/staff UX	C	C	A/R	C	C
Testing/release	C	R	R	A/R	C

Activity	Leader	Backend	Frontend	QA/DevOps	Client
Acceptance/change approval	R	C	C	C	A

12. Budget and cost controls

This is an academic project. Student labour, personal laptops and existing connectivity are in-kind contributions and are not charged to Carpenters Fiji. Their quantities are shown so the budget remains comprehensive. Any cash purchase requires written team/client approval and evidence.

Item	Basis	Cash FJD	Treatment / justification
Student effort	4 x 6 h/week x 14 weeks = 336 hours	0	In-kind academic contribution; track hours for workload control.
Development laptops	Four existing personal devices	0	No new hardware purchase required.
Software/tooling	.NET, React, SQL Server Developer, VS Code, GitHub, Postman, Docker	0	Open-source, developer or existing student access.
Local database	SQL Server container and local storage	0	Development/test only.
Remote staging	Optional short-lived deployment	120	Only if remote client UAT is required and approved.
Email/domain allowance	Optional test sender/domain	50	Supports realistic OTP, notification and invoice testing.
Contingency	Approximately 20% of potential service cost	35	Minor approved hosting/domain variation.
Maximum cash budget		205	Expected actual cash cost may remain zero.

12.1 Budget controls

- Project leader records approval, receipt, purpose and remaining budget.
- No real payment/card service is enabled under this budget.
- Free tiers must still satisfy privacy, reliability and licensing requirements.
- Contingency does not authorise scope expansion.
- Budget status is reviewed at each milestone and final actual cost is reported at closure.

12.2 Economic value rationale

The plan avoids parallel division systems and repeated customer/asset logic. A shared configurable platform can reduce duplicate entry, booking conflicts, unavailable-asset promises, missed service, manual follow-up and reporting effort. Quantified business benefits require client baseline data and are therefore a future business-case activity rather than unsupported savings claims in this academic plan.

13. Risk management

Probability (P) and impact (I) use a 1-5 scale; exposure is P x I. High = 15-25, Medium = 8-14, Low = 1-7. Risks are reviewed weekly and after each client decision.

ID	Risk	P	I	Score	Mitigation / owner
R1	Requirements remain unclear or change late.	4	5	20 H	Time-box discovery, decision log, prototypes and change control / Leader.
R2	Group-wide scope exceeds semester capacity.	4	5	20 H	Protect MVP, capability configuration, MoSCoW priorities / All.
R3	Division/branch authorization exposes data.	3	5	15 H	Server scope queries, negative tests and peer review / Backend + QA.
R4	Booking/status rules allow conflicting or invalid rentals.	3	5	15 H	Transactions, explicit transitions and overlap tests / Backend.
R5	Equipment/operator safety rule is incomplete.	3	5	15 H	Client validation, mandatory gate and no public enablement until accepted / Leader.
R6	Sensitive data or credentials are exposed.	2	5	10 M	Synthetic data, user secrets, review and secret scanning / All.
R7	One member becomes unavailable.	3	4	12 M	Pairing, shared documentation, small reviews and cross-training / Leader.
R8	Integration occurs too late.	3	4	12 M	Continuous integration and fortnightly end-to-end demo / QA.
R9	Environment differences block a member.	3	3	9 M	Docker, Mac/Windows guide and setup verification / QA.
R10	Client availability delays decisions/UAT.	3	4	12 M	Book checkpoints early; focused questions with default/defer date / Liaison.
R11	Email/OTP misconfiguration sends test data incorrectly.	2	4	8 M	Redirect-all staging, delivery log, explicit production approval / Backend.
R12	Data loss or repository damage.	2	5	10 M	GitHub remote, protected review, backup/restore rehearsal / QA.
R13	UI remains too complex for frontline staff.	3	4	12 M	Task language, progressive disclosure, help and representative UAT / Frontend.
R14	Report totals do not reconcile.	3	4	12 M	Defined formulas and source-record reconciliation tests / Backend + QA.
R15	Deployment owner/infrastructure remains unknown.	3	4	12 M	Decision by W11; portable documented fallback / Leader.

13.1 Escalation thresholds

- High risks receive an owner, dated action and immediate leader review.
- Any change affecting scope, milestone, privacy, safety, budget or architecture is escalated to client/supervisor.
- A blocker longer than two working days is raised at the next team contact.
- A suspected privacy/security incident stops affected demonstration/data use until assessed.

14. Governance, communication and change control

14.1 Communication plan

Forum	Frequency	Participants	Output
Team coordination	Twice weekly	Four students	Progress, next actions, blockers and workload.
Iteration planning/review	Every two weeks	Team; client when available	Goal, demonstration, feedback and accepted work.
Client checkpoint	Fortnightly or agreed	Liaison, relevant members, client	Decisions, clarified rules and acceptance notes.
Supervisor update	As scheduled	Team and academic supervisor	Academic guidance and escalations.
Pull-request review	Every functional change	Author plus reviewer	Technical/quality approval and evidence.
Risk/scope review	Weekly	Whole team	Updated risks, decisions and mitigation.

14.2 Change-control process

7. Record the requested change, reason, requester and affected requirement.
8. Estimate value, effort, schedule, cost, security, data, safety and regression impact.
9. Recommend accept, defer, substitute or reject; identify what leaves the MVP if needed.
10. Obtain project leader and client approval for material scope; notify supervisor where assessment/timeline is affected.
11. Update scope, requirements, backlog, schedule, risks, tests and decision log before implementation.

14.3 Decision rights

- Routine implementation within approved acceptance criteria: responsible feature pair.
- Architecture/data changes: backend lead with peer review; leader resolves unresolved delivery trade-off.
- User journey and wording: frontend lead with representative-user/client input.
- Scope, policy, safety, privacy, budget or external commitment: client and project leader; supervisor informed as appropriate.

15. Quality, testing, security and ethics

15.1 Quality plan

Control	Minimum evidence
Build quality	Backend build and frontend TypeScript production build pass.
Review	At least one peer reviews functional changes.
API testing	Positive, validation, unauthenticated, unauthorised, cross-division and cross-branch cases.
Data quality	Required fields, uniqueness, relationships, transitions, overlaps and migration review.
Workflow testing	Vehicle end-to-end; equipment/personnel scenario if enabled; maintenance expense path.
Usability/accessibility	Task completion, responsive widths, keyboard, labels, contrast and plain language.
Security	Authentication, session, recovery/OTP, least privilege, secrets and negative authorization.
Release	Regression, backup/restore, UAT, known issues and deployment rehearsal.

15.2 Test levels and ownership

- Unit/domain tests: booking overlap, transitions, pricing calculations and code verification.
- API integration tests/Postman: endpoints, validation, cookies and role/scope matrix.
- Database/migration tests: clean creation, upgrade, uniqueness, referential integrity and seed repeatability.
- UI tests: primary workflows, error recovery, empty/loading states and responsive layouts.
- Security tests: denied-role and cross-division/branch access, account activation abuse and sensitive-data exposure.
- UAT: client-approved scenarios performed with synthetic representative data.

15.3 Security and privacy controls

- Passwords are handled by ASP.NET Identity hashing; raw passwords are never stored or logged.
- Secrets and bootstrap credentials remain in local/environment secret stores, not Git.
- Staging email defaults to redirect-all until direct delivery is explicitly approved.
- Customer activation and verification codes are expiring, one-time and stored as salted hashes.
- Critical administration, rental and asset changes are auditable.
- Only minimum required information is collected; retention/export/deletion rules require client approval.

15.4 Professional, ethical and safety conduct

The team will protect client confidentiality, distinguish confirmed facts from assumptions, respect software licences, reference external material, report limitations honestly and disclose AI assistance. Safety-sensitive condition and personnel records will not be bypassed or altered without traceability. Decisions balance the client's interests with customer privacy, worker/public safety, accessibility and the wider public interest.

16. Deliverables, deployment and handover

ID	Deliverable	Contents	Acceptance owner
D1	Governance pack	Plan, charter, requirements, risk/decision/change registers and meeting records.	Supervisor / client
D2	Source and database	Reviewed source, migrations, seed guidance, API collection and version history.	Technical representative
D3	CREMS MVP	Responsive application covering accepted modular rental/maintenance workflows.	Client representative
D4	Test and acceptance pack	Cases/results, authorization/security checks, UAT, defects and known issues.	Client + QA lead
D5	Deployment/operations pack	Configuration, deployment, email, backup/restore, monitoring and support notes.	Technical representative
D6	User documentation	Task-based staff guides, customer guidance and administrator instructions.	Client representative
D7	Academic report/demo	Outcomes, evidence, evaluation, reflection and demonstration.	Academic supervisor

16.1 Deployment strategy

- Development: independent local environments using Docker SQL Server and private secrets.
- Staging: client-safe synthetic data, redirect-all email, HTTPS and release candidate configuration.
- Production recommendation: approved host, managed secrets, TLS, backup, logging, restricted administrative access and documented owner.
- Database changes are applied through versioned migrations with backup and rollback decision points.

16.2 Handover checklist

- Release is tagged and reproducible from documented prerequisites.
- Production/staging secrets are supplied through an approved private channel.
- Migration and backup/restore procedures are demonstrated.
- Administrator, manager and rental-officer representatives complete training.
- Known limitations, deferred decisions and future options are documented.
- Acceptance, outstanding actions, support owner and warranty/support period are recorded.

17. Monitoring, reporting and closure

Measure	Target	Review
Milestone completion	All seven accepted or formally re-baselined.	Weekly
Iteration predictability	At least 80% of committed priority items completed.	Fortnightly
Build health	Main/integration branch builds successfully.	Every integration
Defect position	Zero open critical/high defects at handover.	Weekly from W10
Review coverage	Functional changes receive peer review.	Every pull request
Client decisions	Open requests have owner and due date.	Weekly
Team health	Visible workload; no critical knowledge held by one member.	Weekly
Requirements stability	Confirmed/proposed/open status maintained; no unrecorded scope.	Fortnightly

17.1 Weekly status content

- Completed work and evidence links.
- Next-week commitments and owners.
- Milestone confidence: Green, Amber or Red with reason.
- Top risks, defects and blockers.
- Client/supervisor decisions required and due dates.
- Budget/effort variance and team workload concerns.

17.2 Closure criteria

The project closes when accepted MVP scenarios pass; agreed release and documents are handed over; critical/high defects are closed; known limitations are acknowledged; deployment/backup ownership is recorded; the client acceptance position is documented; and the team completes academic submission, demonstration, retrospective and AI-use disclosure.

18. References

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The University of the South Pacific. (2026). CS400 Industry Experience Project, Assignment 1 - Project Plan specification and marking rubric.

Microsoft. (2026). ASP.NET Core documentation: <https://learn.microsoft.com/aspnet/core/> (accessed August 2026).

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React Team. (2026). React documentation: <https://react.dev/> (accessed August 2026).

OWASP Foundation. (2026). OWASP Application Security Verification Standard: <https://owasp.org/www-project-application-security-verification-standard/> (accessed August 2026).

18.1 AI-assisted work declaration

Generative AI assistance was used to help structure this draft, improve wording, analyse planning coverage and generate initial planning artefacts. The student team remains responsible for verifying each statement, rewriting where appropriate, confirming sources and client decisions, replacing placeholders, maintaining confidentiality and documenting AI use in accordance with the assignment specification. AI output is not evidence of client approval and does not replace the team's domain research, professional judgement or authorship.

Appendix A - Team charter

A.1 Purpose

We will work as one accountable team to deliver a secure, usable and evidence-based CREMS MVP for Carpenters Fiji while meeting CS400 academic and professional expectations.

A.2 Team commitments

- Attend agreed meetings or notify the team early when unavailable.
- Complete agreed work by the stated date and raise blockers within two working days.
- Use feature branches, focused commits and peer-reviewed pull requests.
- Review peers constructively and discuss the work rather than the person.
- Protect client information and use synthetic data in development.
- Record material requirements, decisions, changes, risks and evidence.
- Share knowledge through pairing, demonstrations and documentation.
- Report progress and limitations honestly and seek help early.

A.3 Working agreement

Area	Agreement
Primary channels	[INSERT TEAM CHAT] for coordination; GitHub for work/decisions; email for formal client communication.
Response time	Acknowledge team messages within one working day.
Meetings	Twice-weekly team contact and fortnightly planning/review.
Code ownership	Collective ownership; authors obtain peer review before integration.
Definition of ready	User value, acceptance criteria, dependencies, priority and owner are clear.
Definition of done	Meets Section 9.3 and is integrated, tested, reviewed and documented.
File/submission control	Stable filenames; project leader performs final completeness and Moodle submission check.

A.4 Decision-making and conflict resolution

12. Seek consensus using evidence, acceptance criteria, client value and professional obligations.
13. If routine consensus is not reached promptly, the project leader decides after hearing each view.
14. Refer scope, policy, safety, privacy or client-impacting decisions to the client/supervisor.
15. Address interpersonal concerns privately and respectfully; record agreed actions.
16. Escalate unresolved conduct, integrity or persistent workload issues to the academic supervisor.

A.5 Contribution and accountability

Each member plans approximately six project hours per week, adjusted transparently around assessment peaks. Workload is reviewed weekly. Missed commitments are re-planned with an owner and date; repeated issues are discussed with the leader and, if unresolved, the supervisor.

A.6 Charter acceptance

Team member	Student ID	Signature	Date
Kavish Chandra	[INSERT ID]		
[INSERT TEAM MEMBER 2]	[INSERT ID]		
[INSERT TEAM MEMBER 3]	[INSERT ID]		
[INSERT TEAM MEMBER 4]	[INSERT ID]		

Appendix B - Requirements traceability summary

Requirement group	Design / module	Planned evidence
Organisation and scoped access	Division, ServiceOffering, BranchDivision, ApplicationUser scope; API policies/queries	Admin configuration demo and cross-scope negative tests.
Customer identity	Customer portal, registration, verification, staff invitation and account-linked records	Individual/business registration and existing-customer activation tests.
Asset/service catalogue	Asset, division/service links, status, personnel requirement and QR	Catalogue/search, CRUD validation and QR scope tests.
Booking and rental	Booking, item, agreement, inspections, drivers, payments, invoice and lifecycle	Vehicle end-to-end UAT and booking-conflict tests.
Maintenance	MaintenanceJob, status/service dates, expense and dashboard	Return-to-maintenance scenario and cost reconciliation.
Corporate operations	CorporateAccount, quote, approval, job site, PO and dispatch foundation	Corporate scenario subject to approved MVP rules.
Management and audit	Dashboard, reports, management tasks, audit and notification delivery	Reconciliation, audit inspection and email delivery evidence.

Appendix C - Client decisions required

Use this page as a focused agenda for the next requirements meeting. Record the decision, owner and date in the project decision log.

#	Question / decision	Current planning default	Owner / due
1	Which divisions and branches are in the first pilot?	Motors plus one confirmed equipment division; two or more operational branches.	Client / W2
2	Does Carptrac, Hardware or both provide formal equipment hire?	Keep public hire disabled or quote-only until confirmed.	Client / W2
3	Which equipment requires trained personnel and what evidence is mandatory?	Required flag for excavators/backhoes; dispatch gate proposed.	Division representative / before W7
4	What pricing, deposit, cancellation, tax and approval rules apply by division?	Configurable rules; no unsupported production values.	Finance/operations / before W7
5	What is the corporate authorised-contact and purchase-order process?	One account foundation; detailed approval workflow proposed.	Corporate operations / before W7
6	Are property and shipping workflows part of this semester MVP?	Configuration/quote prototype only unless explicitly promoted.	Sponsor / feature baseline
7	What documents, retention periods and privacy notices are required?	Minimum data; synthetic development; seven-year value is not final.	Client/IT/legal / before UAT
8	Who owns deployment, email sender, backup and post-handover support?	Portable staging plan pending technical owner.	Client IT / W11